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The paleomagnetic significance of new U–Pb age data from the Molson dyke swarm, Cauchon Lake area, Manitoba[1]

Henry C. Halls and Larry M. Heaman

Abstract: U–Pb geochronology, paleomagnetism, and petrography indicate that the Molson dyke swarm, along the western margin of the Superior craton, is a composite of at least two ages of intrusion. The more extensive younger dyke set, the Molson swarm sensu stricto, generally has a 030° trend, is mainly pyroxenitic to noritic with subordinate diabase, and has been related to rifting in a back-arc environment during closure of the Manikewan ocean at about 1920–1800 Ma which culminated in the Trans-Hudson Orogen. A U–Pb zircon age from one of these dykes, located at Cauchon Lake, indicates emplacement at 1877^{+7}_{-4}, similar to two previous U–Pb age determinations on Molson dykes. Another dyke from Cauchon Lake yields a baddeleyite–zircon U–Pb date of 2091 ± 2 Ma and appears to be part of an older, mainly diabasic suite of east-northeast-trending dykes that may represent a continental rifting episode that preceded the opening of the Manikewan ocean. The new U–Pb age data require a revision to the interpretation of the A, B, and C paleomagnetic poles previously reported from Molson dykes. The A pole (16.1°N, 96.5°W), initially assigned an age of 1883 Ma, is now considered to be younger and derived from a Paleoproterozoic overprint associated with the Trans-Hudson Orogen at about 1700–1800 Ma. Pole B (27.1°N, 140.8°W) from the Molson swarm sensu stricto is now regarded as primary, and dated at 1880 Ma. Pole C can be subdivided into two poles, one virtually the same as B but of opposite polarity (and therefore about 1880 Ma old) and a new pole (53°N, 180°W) derived from a primary remanence and dated at 2091 Ma. The new paleomagnetic interpretations may have important consequences for tectonic models of the Trans-Hudson Orogen and for Paleoproterozoic continental reconstructions.

Résumé : La géochronologie U–Pb, le paléomagnétisme et la pétrographie indiquent que l'essaim de dykes Molson, le long de la bordure ouest du craton du Supérieur, est composé d'au moins deux âges intrusifs. L'ensemble de dykes plus étendu et plus jeune, l'essaim Molson sensu stricto, a généralement une tendance 030°; il a une composition pyroxénitique à noritique et une diabase accessoire. On l'a relié a une zone de fracture dans un environnement d'arrière arc lors de la fermeture de l'océan Manikewan il y a environ 1920–1800 Ma et qui a culminé avec l'Orogène transhudsonien. Une datation U–Pb sur du zircon provenant de l'un de ces dykes localisé au lac Cauchon indique une mise en place il y a 1877^{+7}_{-4} Ma, ce qui est semblable à deux déterminations antérieures d'âge sur des dykes Molson. Un autre dyke du lac Cauchon donne une date U–Pb baddeleyite–zircon de 2091 ± 2 Ma et semble faire partie d'un ensemble de dykes où domine le diabase et dont la tendance est ENE; ces dykes pourraient représenter un épisode de distension du continent précédant l'ouverture de l'océan Manikewan. Les nouvelles données d'âges U–Pb exigent une révision de l'interprétation des pôles paléomagnétiques A, B et C antérieurement tirés des dykes Molson. Le pôle A (16,1°N, 96,5°W), pour lequel on a antérieurement déterminé un âge de 1883 Ma, est maintenant considéré comme étant plus jeune et provenant d'une surimpression paléoprotérozoïque associée à l'Orogène trans-hudsonien il y a environ 1700–1800 Ma. Le pôle B (27,1°N, 140,8°W) de l'essaim de dykes Molson sensu stricto est maintenant considéré comme primaire et âgé de 1880 Ma. Le pôle C peut être subdivisé en deux pôles, l'un pratiquement identique à B mais de polarité inverse (et donc vieux d'environ 1880 Ma) et un nouveau pôle (53°N, 180°W) dérivé d'une rémanence primaire et datant d'environ 2091 Ma. Les nouvelles interprétations paléomagnétiques peuvent avoir des conséquences importantes pour les modèles tectoniques de l'Orogène trans-hudsonien et pour la reconstruction continentale au Paléoprotérozoïque.

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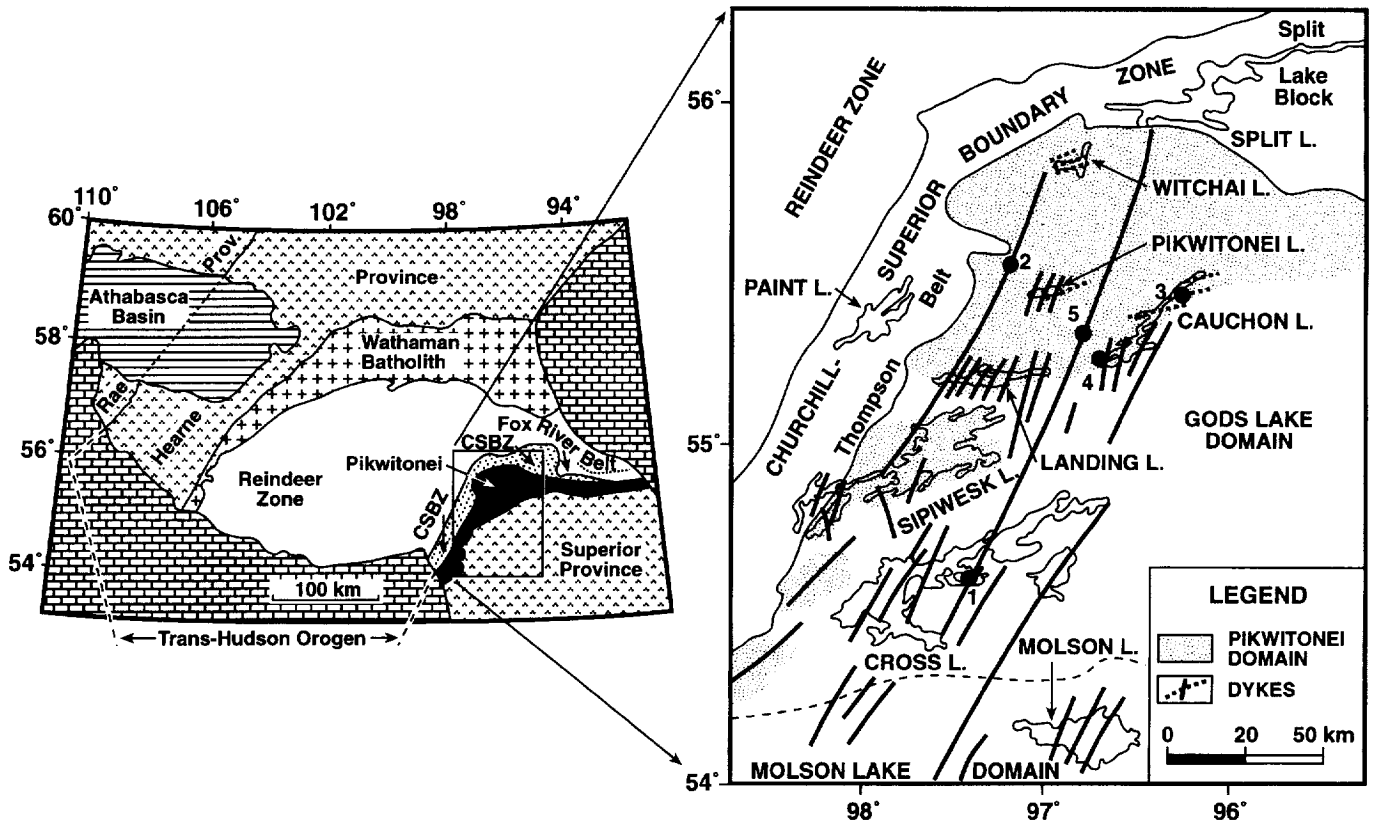
Introduction

The Molson dyke swarm, located on the western margin of the Superior Province in Manitoba (Fig. 1), has been the subject of several radiometric and paleomagnetic investigations because it has long been considered the remnant of a Paleoproterozoic rift-parallel magmatic event that heralded the opening of the Manikewan ocean (Stauffer 1984). This ocean was formed by rifting of an Archean landmass, and its subsequent closure, first expressed by island-arc magmatism

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Fig. 1. General geological map of the Molson dyke swarm (after Scoates and Macek 1978), showing its location with respect to the Trans-Hudson Orogen, the distribution of Molson dykes sensu stricto (solid lines), and older dykes (broken lines). Numbered locations are as follows: 1 and 2, Cross Lake and Cuthbert Lake dykes, respectively, sampled for U–Pb dating by Heaman et al. (1986); 3 and 4, U–Pb/paleomagnetic sites MO91 and MO77, respectively; 5, paleomagnetic site MO89 on the Nelson River dyke. CSBZ, Churchill–Superior boundary zone.



at 1.92 Ga (Gordon et al. 1990), ultimately formed the ~1800 Ma Trans-Hudson Orogen (THO). A previous U–Pb geochronology project was initiated by Heaman et al. (1986) to establish the emplacement age of the Molson dykes, thereby constraining the timing of the rifting event. Identical high-precision U–Pb zircon ages for two dykes (locations 1 and 2 in Fig. 1) at Cross Lake (1884 ± 2 Ma) and Cuthbert Lake (1883 ± 2 Ma) were interpreted to indicate that most, if not all, of the Molson magmatism was younger than island-arc volcanism occurring within the Manikewan ocean (Heaman et al. 1986), thus demonstrating that the majority of the Molson dykes were not related to the pre-1.92 Ga rifting event.

To resolve this paradox, further research was carried out to determine whether there were indeed multiple ages of dyke emplacement, a suggestion made by Corkery (1985) and supported by crosscutting relationships between mafic dykes observed in the Split Lake block (Fig. 1). Two additional lines of evidence have been put forward that support the field observations of Corkery: (*i*) a paleomagnetic study of the Molson swarm (Zhai et al. 1994) identified three magnetization directions, designated A, B, and C, all of which were regarded as primary and having distinct ages; and (*ii*) a U–Pb zircon age of 2072 ± 3 Ma was obtained for a northeast-trending gabbro dyke at Birthday Rapids in the Split Lake block (Heaman and Corkery 1996).

Component A was assigned an age of 1883 Ma by Zhai et al. (1994) on the basis of aeromagnetic anomaly correlation of the dated Cuthbert dyke with A component dykes identified at Landing Lake to the south (Fig. 1). Based on a comparison with well-constrained paleomagnetic pole positions from the Superior Province (e.g., Buchan et al. 1993), Zhai et al. estimated the emplacement age of the B and C dykes to be between 2170 and 2120 Ma.

Here we discuss the paleomagnetic and age implications for the Molson swarm in terms of its origin and relation to the THO in light of new zircon–baddeleyite U–Pb ages obtained for two diabase dykes located along the shore of Cauchon Lake (Fig. 1).

Geological background

The approximately north-northeast-trending Molson dyke swarm occurs within a 100 km wide corridor along the western margin of the Archean Superior Province (Fig. 1). Abundant well-preserved Molson dykes transect the high-grade granulites of the Pikwitonei Domain and are observed to be rotated and metamorphosed within the Thompson belt (Fig. 1) where migmatic gneisses, largely derived from Pikwitonei granulites, were formed through reworking by the THO (Weber 1990). Within the corridor, the Molson dyke swarm covers an area of approximately 26 000 km^2

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and represents a volume of magma of the order of 50 000 km^3.

Molson dykes are up to 60 m in width and vary in rock type from gabbro to pyroxenite. Their composition ranges from komatiitic basalt through to more normal tholeiite, the latter in particular exhibiting well-developed chilled margins. A detailed report on the geology and geochemistry of the Molson dykes was published by Scoates and Macek (1978).

U–Pb geochronology

Geology and paleomagnetism of samples

The samples used in this study were collected from pegmatitic pods within two dykes located at Cauchon Lake (Fig. 1) and are from paleomagnetic sites MO91 and MO77 of Zhai et al. (1994). The dyke at site MO91 (location 3 in Fig. 1) has a width of 36 m, trends 060°, and dips 75°NW, approximately conformable with the gneissosity of adjacent host rocks. Petrographically, the dyke is fresh from chilled margin to interior and comprises subophitic diabase with equal proportions of clinopyroxene and plagioclase. Paleomagnetically the site gave a well-defined C magnetization (declination (D) = 120°, inclination (I) = –59°, radius of 95% confidence circle about the direction (α_{95}) = 5.5°, number of oriented cores (N_S) = 7, from both interior and western chilled margin).

Site MO77 (location 4 in Fig. 1) is from an outcrop located in the interior of a fresh dyke composed of plagioclase, clinopyroxene, orthopyroxene, olivine, brown amphibole, and accessory biotite and magnetite. This dyke is coarse grained, contains abundant pegmatite, and is at least 13 m wide with unexposed margins. Faint internal banding suggests a trend of about 070°. Paleomagnetically, the dyke records a well-defined B direction (D = 260°, I = 67°, α_{95} = 2.7°, N_S = 6).

Analytical techniques

For both samples, approximately 1 kg of drill core was crushed to a fine powder (<300 μm) using a jaw crusher and Bico Disk mill. Baddeleyite–zircon isolation was accomplished using a Wilfley table, combined with standard heavy mineral and magnetic separation techniques. A small amount of both baddeleyite and skeletal zircon shards was recovered from dyke MO91, with zircon tending to form larger (up to 100 μm) and more abundant grains than baddeleyite. Most of the baddeleyite grains were altered, with few pristine crystals present. Sample MO77 contained abundant zircon and pyrite. The zircon varied in colour from colourless to tan, and generally formed irregular fragments. A few broken long prismatic grains were present and typically contained longitudinal inclusion trains. The incipient development of a younger polycrystalline zircon growth was detected on some zircon fragments. Also present were a number of elliptical resorbed zircon grains interpreted to be xenocrystic and not selected for analysis.

Isolation of U and Pb by anion exchange chromatography and determination of their isotopic composition on a VG354 mass spectrometer, operated in single-collector mode, generally followed procedures outlined elsewhere (Heaman and Machado 1992; Heaman and LeCheminant 1993). For sample MO91 two analyses were determined at the University of Alberta, and three at the Jack Satterly Laboratory, Royal Ontario Museum (Table 1). All zircon U–Pb analyses for dyke MO77 were carried out at the Radiogenic Isotope Facility, University of Alberta. At the time of this study (1993–1996), the analytical blanks for U and Pb measured in the two laboratories were 0.5 and 2.0 pg (Royal Ontario Museum) and 1 and 6 pg (University of Alberta), respectively. Age calculations and associated uncertainties were calculated using the program of Davis (1982) and decay constants 1.55125×10^{-10} year^{-1} for ^{238}U and 9.8485×10^{-10} year^{-1} for ^{235}U as determined by Jaffey et al. (1971).

U–Pb results

Dyke MO91

The U–Pb results for five fractions (three zircon and two baddeleyite) are presented in Table 1 and on a concordia diagram in Fig. 2*a*. Two of the zircon fractions (3 and 4) were given an abrasion treatment (Krogh 1982) and plot within error of the concordia curve with ^{207}Pb/^{206}Pb model ages of 2091 and 2095 Ma, respectively. The two baddeleyite fractions (2 and 5; solid ellipses in Fig. 2*a*) are significantly more discordant than the zircon analyses and may have been affected by Pb loss associated with the pervasive alteration of the grains. An interesting feature of these baddeleyite grains is their higher than normal thorium contents (note that the Th/U ratios in Table 1 are much higher than 0.1, a maximum value for typical baddeleyite). Baddeleyite analysis 5 represents the best attempt at isolating the least altered grains. Regression of four of the five multigrain analyses (excluding 2, the most obviously altered baddeleyite grains) yields an upper intercept age of $2091.1^{+1.8}_{-2.1}$, interpreted to be the emplacement age of this dyke. This is the oldest emplacement age yet recorded for a "Molson" dyke.

Dyke MO77

The U–Pb results for three multigrain zircon fractions (11–61 grains) are presented in Table 1 and displayed on a concordia diagram in Fig. 2*b*. Although the three analyses are not collinear, two (1 and 3) have nearly identical ^{207}Pb/^{206}Pb ages of 1875 and 1874 Ma, respectively. The reference line constructed to pass through these two analyses yields an upper intercept age of 1877^{+7}_{-4}, which we interpret as close to the emplacement age for this dyke. Analysis 2 has a slightly older ^{207}Pb/^{206} Pb age of 1887 Ma and significantly lower uranium content and Th/U ratio (see Table 1). It is possible that a small number of inherited grains were unknowingly included in this fraction.

Paleomagnetic implications

An examination of the apparent polar wander path for the southern Superior Province (Fig. 3), based on precisely U–Pb dated dyke swarms, shows that pole positions corresponding to the 2214 Ma Senneterre dykes, 2167 Ma Biscotasing dykes (Buchan et al. 1993), and 2121 Ma Marathon Normal (N) and 2076 Ma Fort Frances dykes (Halls 1986; Buchan et al. 1996*a*) form a northwest-trending path, with the poles lying in chronological order. The Molson C pole, dated here at 2091 Ma, lies between the Biscotasing

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Table 1. U–Pb zircon–baddeleyite results for two Cauchon Lake diabase dykes, Manitoba.

	Weight (μg)[a]	U (ppm)	Th (ppm)[b]	Pb (ppm)	Th/U	TCPb (pg)[c]	$^{206}Pb/^{204}Pb$[d]	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	Model ages (Ma) $^{206}Pb/^{238}U$	Model ages (Ma) $^{207}Pb/^{235}U$	Model ages (Ma) $^{207}Pb/^{206}Pb$	%Disc.[e]
MO77														
1Z	40(11)	314.1	702.6	160.0	2.2	32	8 330	0.3302±6	5.222±11	0.11469±6	1839.5±3.0	1856.2±1.8	1875.0±1.0	2.2
2Z	21(23)	70.3	120.2	33.0	1.7	15	2 070	0.3299±6	5.251±11	0.11544±9	1838.0±2.9	1861.0±1.8	1886.8±1.5	3.0
3Z	31(61)	625.4	1468.6	319.6	2.4	56	6 911	0.3248±6	5.133±10	0.11461±6	1813.2±2.8	1841.5±1.7	1873.7±1.0	3.7
MO91														
1Z[f]	14(19)	256.0	209.4	114.0	0.8	4	23 538	0.3773±5	6.727±9	0.12933±9	2063.4±2.4	2076.2±1.2	2088.9±1.2	1.4
2B[f]	7(22)	352.1	223.8	145.7	0.6	8	7 492	0.3636±6	6.443±11	0.12853±10	1999.1±2.8	2038.2±1.4	2077.9±1.3	4.4
3Z[f]	6(16)	210.9	162.3	94.9	0.8	3	8 812	0.3846±5	6.867±10	0.12950±9	2097.8±2.5	2094.5±1.3	2091.2±1.2	–0.4
4Z	2(50)	300.6	260.6	146.5	0.9	27	543	0.3819±8	6.831±20	0.12975±24	2084.9±3.9	2089.8±2.7	2094.6±3.3	0.5
5B	14(76)	232.4	87.0	94.2	0.4	34	2 275	0.3731±7	6.657±14	0.12943±11	2043.7±3.3	2067.0±1.9	2090.2±1.4	2.6

Note: All atomic ratios corrected for fractionation, common Pb (Stacey and Kramers 1975), and blank Pb (2 pg) / U(0.5 pg). Ages calculated using the following decay constants determined by Jaffey et al. (1971): 1.55125×10^{-10} year^{-1} for ^{238}U and 9.8485×10^{-10} year^{-1} for ^{235}U. Regression treatment follows the method of Davis (1982). B, baddeleyite; Z, zircon.

[a]Number in parentheses refers to the number of grains analyzed.
[b]Estimated from $^{208}Pb/^{206}Pb$ ratio and $^{207}Pb/^{206}Pb$ age.
[c]Total common Pb present in the analysis.
[d]Corrected for fractionation and spike contribution.
[e]Percent discordance from $^{207}Pb/^{206}Pb$ age.
[f]Determined at the Jack Satterly Laboratory, Royal Ontario Museum.

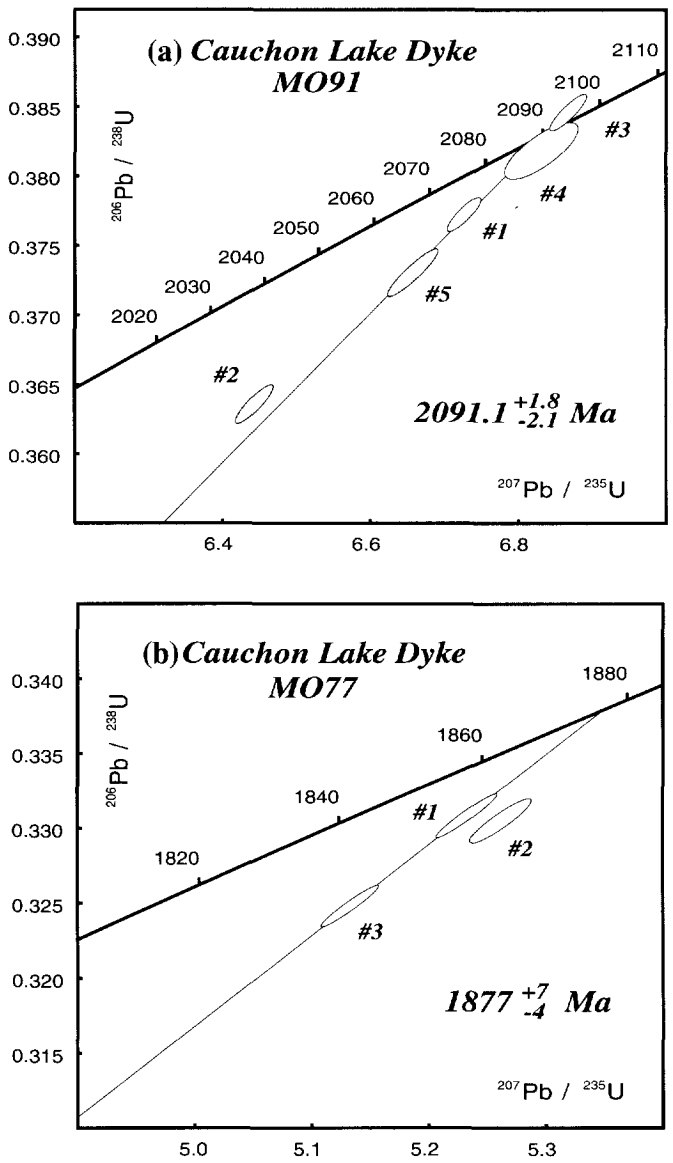


Fig. 2. U–Pb concordia plots for two dykes from Cauchon Lake: (*a*) MO91, and (*b*) MO77. Open and solid ellipses denote zircon and baddeleyite fractions, respectively.

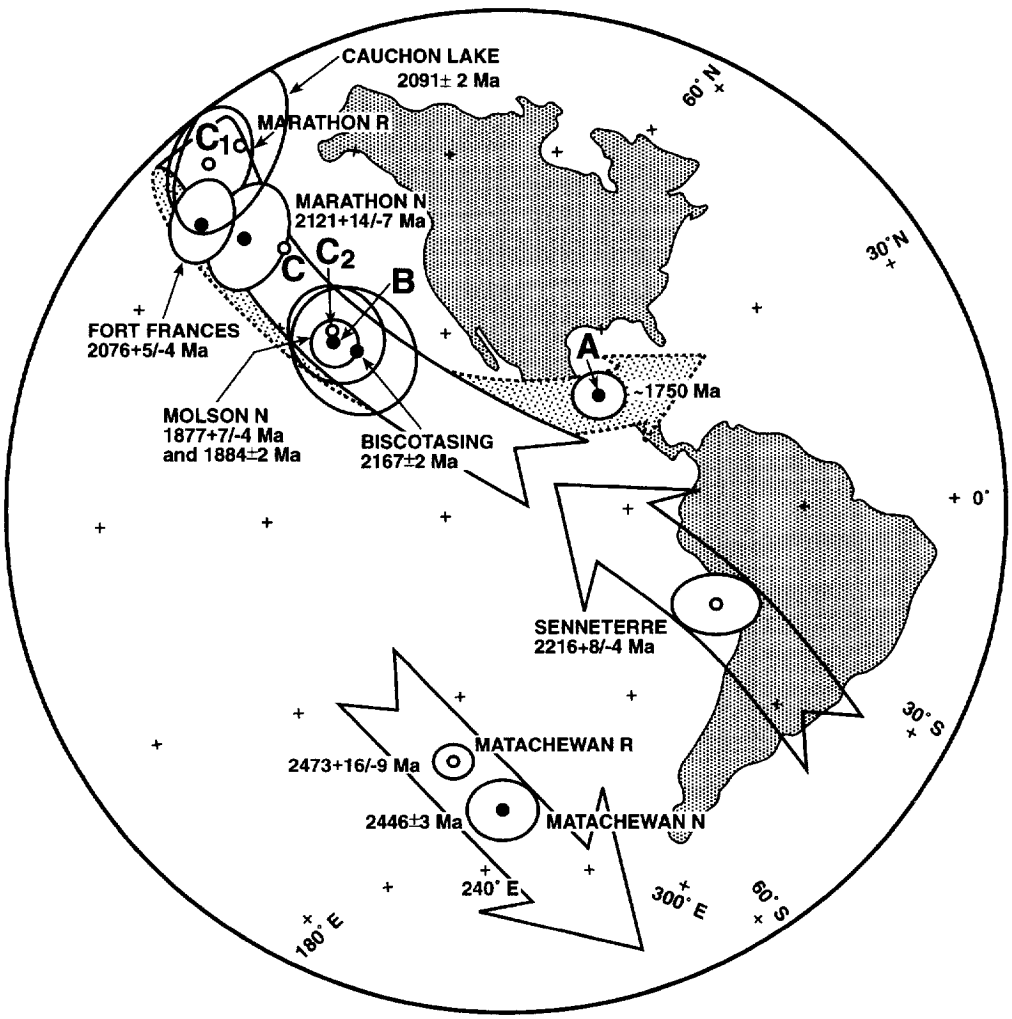
and Marathon N poles (Fig. 3) and therefore appears out of sequence. This observation and the streaked distribution of site directions defining pole C (Fig. 4) prompted a reexamination of the existing paleomagnetic data.

Additional thin sections were made so that all dykes having C magnetizations could be examined petrographically. Two groups of dykes can be separated on the basis of petrographic appearance, trend, and geographic location (Fig. 4). A northern group, composed exclusively of ophitic to subophitic diabase with a generally east-northeast trend, includes the 2091 Ma dyke investigated in this study and records a paleomagnetic pole position C_1 (latitude 53°N; longitude 180°W; d_p = 7.6° and d_m = 9.7°, where d_p and d_m are the semi-minor and semi-major axes, respectively, of the 95% confidence ellipse about the pole; and number of dykes (N_D) = 8). This pole is significantly different at the 95% confidence level from pole C_2 (28.7°N, 144°W, d_p = 7.3°, d_m =

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Fig. 3. Apparent polar wander path for the southern Superior Province for the time period 2470 to ~1750 Ma. Only poles from Paleoproterozoic dykes are shown with their 95% confidence ovals and assigned U–Pb ages. Paleomagnetic field tests show primary remanences for the Matachewan, Senneterre, Biscotasing, Marathon Reverse polarity, Molson, and Fort Frances swarms (Buchan et al. 1993, 1996*a*). Poles labelled A, B, and C are the original ones from the Molson dyke swarm found by Zhai et al. (1994). Solid circles represent pole positions calculated from the measured magnetization directions, and open circles after directions have been antipodally reversed. The apparent polar wander path is drawn with its stippled portion representing a southeast-directed segment, although by plotting the A and B poles antipodally the polar wander path could continue around to the obverse side of the Earth to that shown and would reverse direction at about 1900 Ma. U–Pb age data are from Heaman (1997) prior to 2400 Ma; Buchan et al. (1993, 1996*a*) for the time period 2200–2076 Ma, with the exception of the Cauchon Lake pole (this paper); and Heaman et al. (1986) and this paper for poles younger than 1900 Ma.



9.1°, N_D = 11) given by a geographically more widespread dyke set that trends north-northwest to north-northeast and has a mainly pyroxenite to norite (±olivine) composition with subordinate diabase.

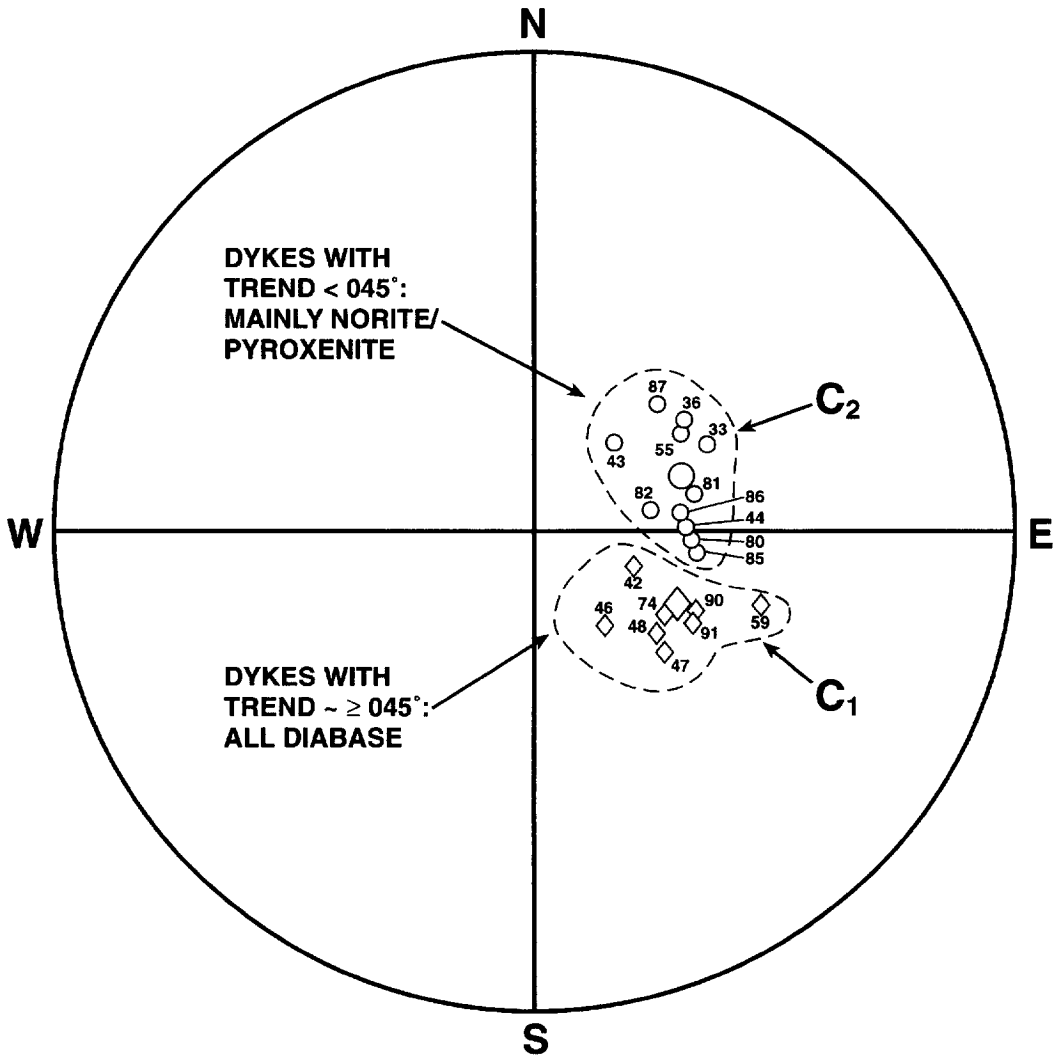
The C magnetization was originally considered primary because unbaked granulite gneisses collected from the western part of Pikwitonei Lake and in the central one-third of Landing Lake (Fig. 1), where both B and C dykes are found, gave a distinctly different direction D (D = 53.6°, I = –64.4°) that was regarded as older (Zhai et al. 1994). However, now that C has been resolved into C_1 and C_2, evidence for primary magnetization for C_2 is lacking. The D magnetization is only about 10° from C_2, and both are virtually antipodal to B. Positive baked contact tests indicate that B is primary (Fig. 7 of Zhai et al. 1994) and the new U–Pb data from site MO77 indicate that it is about 1880 Ma old. Therefore, D and C_2 may also date approximately from this time. The C_1 pole occurs northwest of that for the 2121 Ma Marathon N dykes (42.8°N, 164.4°W), consistent with its assigned age of 2091 Ma.

The primary nature of C_1 can be demonstrated from site MO42 (D = 112°, I = –73°, α_{95} = 6, N_S = 7; Zhai et al. 1994) where baked granulite adjacent to the dyke yields the same remanence direction as the dyke, whereas unbaked granulite almost three dyke widths away gives a high coercivity component with a significantly different direction (Fig. 5). The C_1 dykes are restricted to the northern part of the area (Fig. 1). The occurrence of the 1880 Ma B pole to the southeast of the 2076 Ma Fort Frances dyke pole (42.7°N, 175.8°W) and the 2091 Ma pole C_1 (Fig. 3) sug-

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Fig. 4. Equal-area stereonet subdividing the original streaked distribution of paleomagnetic directions defining the C magnetization (see Table 1 and Fig. 6*a* of Zhai et al. 1994) into two clusters, based on the petrography and trend of the contributing dykes, that give mean directions C_1 and C_2. Open symbols indicate upward-pointing magnetizations, the mean of which is given by the larger symbol for each cluster. Numbers beside points are the MO site numbers from Zhai et al. (1994).



gests, assuming minimum path length and therefore rejection of the antipodal position of the B pole, that by about 2080 Ma the polar track had reversed direction to move back towards the southeast and through the 1880 Ma B pole (Fig. 3).

The antipode of the undated reversed C_2 pole position is virtually coincident with that given by B dykes (27.1°N, 140.8°W) which also include both olivine-bearing pyroxenitic–noritic and diabasic types. The two petrographically similar dyke suites will collectively be referred to as the Molson swarm sensu stricto. The occurrence of C_2 in the dykes and pervasively, as D, in the granulites suggests that the Molson dyke swarm was intruded during a time interval that included a reversal of the Earth's field direction from C_2 to B.

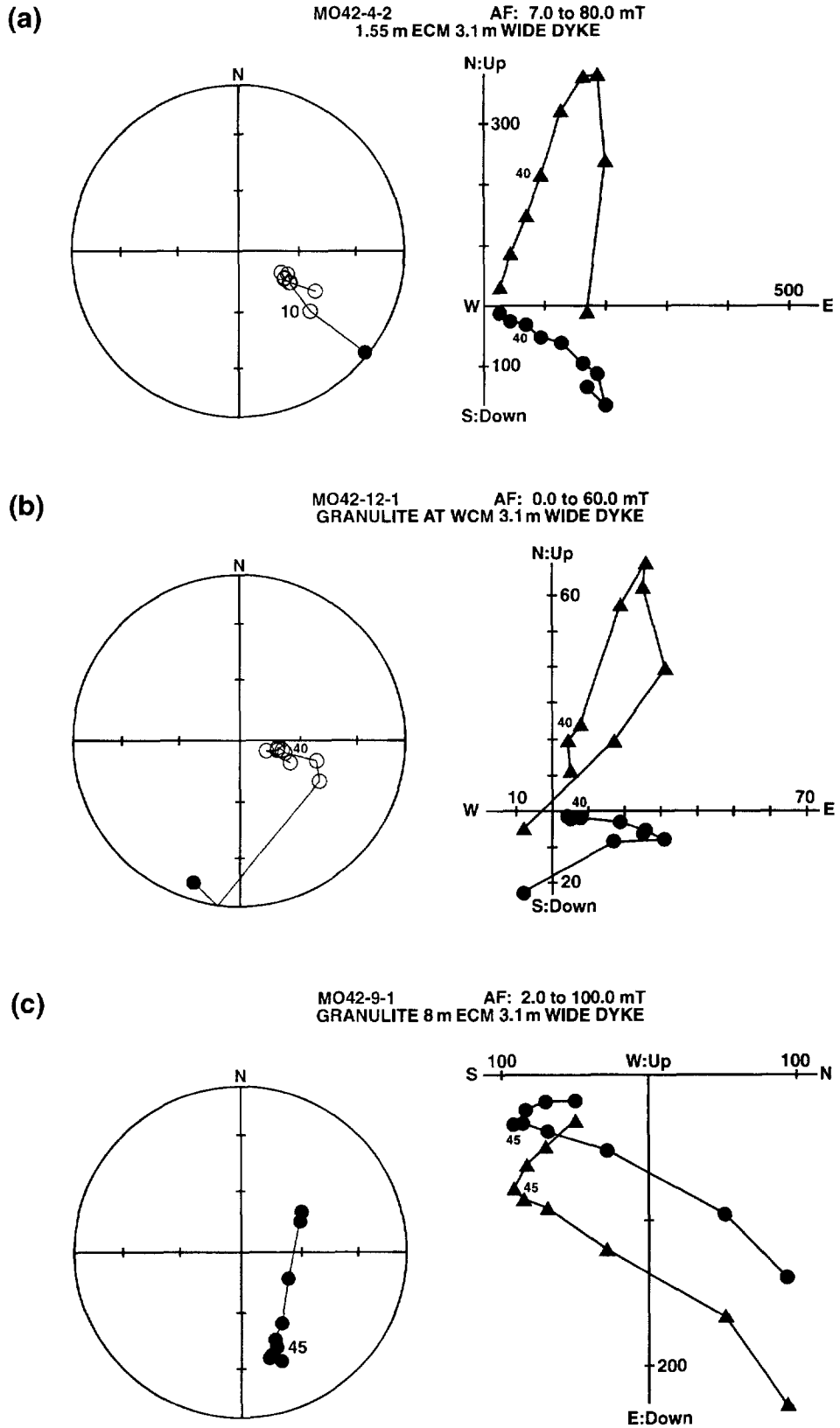
The new age for the B pole is in direct conflict with that reported earlier for the A component (Zhai et al. 1994), which was assigned an age of 1883 Ma on the strength of the Cuthbert dyke U–Pb data (Heaman et al. 1986) and possible correlations of this dyke with those in Landing Lake which carry A. Four lines of evidence suggest that B is indeed about 1880 Ma and, moreover, that A is younger.

First, a dyke, about 100 m wide, follows the Nelson River and is known as the Nelson River dyke. It is the eastern of two megadykes in the Molson swarm that are known to extend for several hundred kilometres (Fig. 1). The dyke is correlated with one in Cross Lake that has given a U–Pb age of 1884 ± 2 Ma (Heaman et al. 1986). The Nelson River dyke has an ultramafic composition and is only exposed at low water levels because it has been preferentially eroded to form the river channel. Site MO89 (location 5 in Fig. 1) of Zhai et al. (1994) is a vertical, 3 m wide diabase dyke, trending 022°, parallel to the river on its western side (Fig. 1). It gives a B magnetization (D = 227°, I = 63°, α_{95} = 5.8°, N_S = 7). This dyke is unlikely to postdate the Nelson dyke because in the general area of the Molson dykes mafic magmatism younger than 1880 Ma is unknown. Therefore, it is either an apophesis of the Nelson River dyke or, if older, would lie well within the baked zone of the larger dyke and thus would have been magnetically reset by it.

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Fig. 5. Paleomagnetic results, after alternating field demagnetization, from a 3.1 m wide dyke and host rocks at site MO42 from Witchai lake, plotted on equal-area stereonets and corresponding vector diagrams. (*a*) Dyke interior. (*b*) Baked granulite adjacent to the western chilled margin (WCM). (*c*) Unbaked granulite 8 m from the eastern chilled margin (ECM). On the stereonets solid and open circles refer to downward- and upward-directed magnetizations, respectively. In the vector diagrams, solid circles and triangles represent projections of the tip of the magnetization vector on the horizontal and vertical planes, respectively. Magnetization intensity units are $A \cdot m^{-1} \times 10^{-3}$.



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The foregoing evidence suggests that the 1884 Ma Nelson River dyke was intruded at a time when the Earth's magnetic field had a B direction.

Second, the A magnetization was assigned to the 1883 Ma Cuthbert dyke on the basis of a possible correlation with A dykes at the western end of Landing Lake using regional aeromagnetic data. The correlation is uncertain because the linear "dyke" anomalies are not continuous and the dated Cuthbert dyke could pass to the west of Landing Lake (see Fig. 8 of Zhai et al. 1994).

Third, three A dykes at sites MO32, MO51, and MO64 have paleomagnetic results that are consistent with a primary origin for A, because unbaked host rocks in their vicinity give D magnetizations (Zhai et al. 1994). Others, such as those at the western end of Landing Lake, have no host rock data in their vicinity and, therefore, their magnetization could be secondary. The greater abundance of A dykes with increasing proximity to the THO (Fig. 5 of Zhai et al. 1994) supports a secondary origin for A.

Fourth, the secondary nature of A is more strongly suspected for the majority of A dykes found in the Sipiwesk Lake area (Fig. 5 of Zhai et al. 1994), which is closest to the Thompson belt. Here the dykes locally have a foliation parallel to enclosing gneisses and have been recrystallized with the production of secondary amphibole (sites MO14 and MO16). Most of the dykes (all except sites MO21 and MO22) are cut by numerous thin felsic pegmatite veins and frequently have sheared chilled margins. Unfortunately, all granulite gneisses remote from dykes (sites MO18, MO23, and MO24) were paleomagnetically unstable, and thus a negative baked contact test, necessary to prove the secondary origin for A, is lacking. However, the field observations are compelling; dykes showing these secondary features generally do not yield primary magnetizations. For those A dykes (MO32, MO51, and MO64) more remote from the adjacent Thompson belt, where nearby dykes carry B or C, further tests are needed to demonstrate if the A remanence is primary and thus indicative of a still younger magmatic event.

From the foregoing discussion, the age of the A pole is younger than the primary remanence of the 1880 Ma Molson dykes. The dykes have been deformed, presumably in the final collisional–magmatic phases of the THO from about 1720 to 1820 Ma (Machado 1990) and would have acquired their remanence after subsequent cooling. Therefore, without Ar–Ar data, the age of the A pole can only be defined within broad limits of about 1.7–1.8 Ga. The A pole position thus continues the apparent polar wander path, as shown in Fig. 3, on its southeast-directed return journey, although an excursion to higher latitudes, either before or after the A magnetization was formed, is indicated from paleomagnetic studies on units within the orogen which are described in an accompanying paper (Symons and Harris 2000).

Discussion and conclusions

The new U–Pb age data prove conclusively that the Molson dyke swarm sensu lato is a composite of at least two episodes of dyke intrusion separated in time by about 200 Ma. The older, 2091 Ma dykes occur in the northern part of the swarm, are diabasic, and trend generally east-northeast (Fig. 1). Their age is only slightly younger than the zircon–baddeleyite age of 2111.2 ± 0.6 Ma obtained from a rift-related gabbro in the lower Hurwitz Group of the Hearne Province (Heaman and LeCheminant 1993). The similarity in age and location of these mafic magmatic events, on opposite sides of the THO, suggests that they are related to the opening of the Manikewan ocean (Halls and Heaman 1997). This in turn implies that the Hearne and Superior Provinces (Fig. 1) were once contiguous before they rifted apart.

The younger ~1880 Ma dykes, which make up the Molson swarm sensu stricto (Fig. 1), tend to be more ultramafic in composition, generally have north-northeast trends (although north, northwest, and east-northeast dykes are known), and may have originated as a result of rifting in a back-arc setting (Weber 1990).

For the time interval 2230 to ~1750 Ma the paleomagnetic apparent polar wander path for the southern Superior Province, based entirely on mafic dykes, is shown in Fig. 3. A similar path, using mafic dykes and associated volcanics, has been obtained from the Minto Block in the extreme northeastern part of the Superior Province (Buchan et al. 1998), although, like the path in Fig. 3, it is poorly dated at its younger end. In this study, a key pole (38.2°N, 186.2°W), from the Minto dykes (U–Pb age on zircon and baddeleyite of 1998 ± 2 Ma), plots in the cluster of poles formed by the Fort Frances, Marathon, and C_1 dykes in Fig. 3. Incorporating both data sets in the Superior Province apparent polar wander path suggests that from ~2200 to ~2100 Ma the pole moved about 1° arc per Ma towards the northwest, and then for the next 100 Ma, from 2100 to 2000 Ma, was more or less stationary. From ~2000 to ~1750 Ma, the rate of polar wander depends upon the choice of antipodal pole positions for the A and B poles. If the antipode of B is chosen, maximizing the path length with respect to the cluster at 2100–2000 Ma, the polar wander path (version 1) resumes its fast rate of movement of about 1° arc per Ma, in the same sense as before. If the antipode of A is also plotted, the path reverses its motion to form a hairpin (Irving and Park 1972) at about ~1900 Ma. If both the A and B poles are plotted as in Fig. 3, the path (version 2) has a hairpin with an apex defined at about 2100–2000 Ma and a rate of polar motion along the return path of only about 0.4° per Ma. Although version 1 of the path violates the minimum path length concept, it yields a hairpin with an apex corresponding closely with the time of ocean closure that culminated in the THO. The meaning of hairpins in terms of changes in the direction of plate motion associated with Wilson cycles and more specifically with orogens was first suggested by Irving and Park (1972). Since the polar wander path of version 2 (Fig. 3) remains within about 30° of arc from the sampling sites throughout the time period 2167 to ~1750 Ma, the associated hairpin cannot be readily interpreted in terms of a reversal of plate motion, and the age of its apex (about 2000–2100 Ma) does not correspond with any known orogen.

The apparent polar wander path for the time period 1700–1900 Ma is extremely uncertain because poles determined from reliably dated remanences are absent. Symons and Harris (2000) suggest that a case can be made for an ~1830 Ma hairpin that corresponds to orogenic collision of juvenile terranes of the THO. Their poles are derived from different units within the THO, mainly gneisses, granites, and gab-

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bros, that yield zircon and monazite ages between 1890 and 1770 Ma, which are thought to represent the approximate time of remanence acquisition (Symons et al. 1995; Symons and Harris 2000) The A pole which may also be related to the final phase of the orogen, dated by U–Pb on late pegmatites and gneisses in the adjoining Thompson belt to be from 1720 to 1820 Ma (Machado 1990), appears to be in conflict with most of the THO data, as it plots about 30° of latitude farther south. The problem is that the closure temperatures for Pb in zircon and monazite (>800°C and ~725°C, respectively; Heaman and Parrish 1991) are much higher than those temperatures at which a remanence carried by magnetite is acquired (550–580°C), assuming it is entirely a thermal remanence, a difference that is even greater if remanences have a significant chemical component.

A U–Pb study of zircon, monazite, and titanite from the Hearne Province, which forms the westernmost domain of the THO (Fig. 1), documents a high-grade thermotectonic event at about 1810–1830 Ma, with maximum temperatures of ~750°C and a cooling rate of ~7°C/Ma (Orrell et al. 1999). These authors suggest that this high-temperature regional thermotectonism, which was associated with terminal collision, extended across the Trans-Hudson Orogen to the Thompson Fold Belt in the west. If this conclusion is valid, any thermal remanences formed during subsequent uplift and cooling from within the THO itself could be at least 25 Ma younger than associated U–Pb zircon and monazite ages. Therefore the poles reported from within the orogen (Symons and Harris 2000) have ages that are uncertain with respect to the A pole and have been omitted from Fig. 3.

Despite the uncertainties in the form and interpretation of the Superior Province apparent polar wander path, this work has resulted in a significant change by about 40° of arc in the 1880 Ma pole position. The 2000–1750 Ma segment that includes this data point has been used previously to help fit Laurentia to Baltica at about 1.9 Ga (Buchan et al. 1996*b*)and to chart the evolution of the THO (Symons et al. 1995), particularly with respect to how the Superior and Hearne margins of the orogen behaved during subduction and in the early stages of continental collision. These interpretations will require modification arising from the repositioning of the key 1880 Ma pole (e.g., Symons and Harris 2000).

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